

Technical and Business Writing

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Technical Proposal

Types of Technical Proposal

Types of Technical Proposals

Classification of the proposals

Who asked for it?

Initiation-based

Who is it for?

Audience-based

Why is it written?

Purpose-based

Who asked for it?

Solicited Proposal

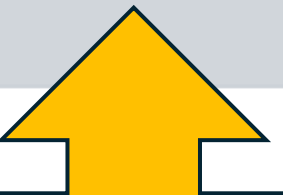
Written in response to a formal Request for Proposal (RFP). The problem is already defined; your job is to respond with the best solution.



Example: A university invites companies to propose a Learning Management System

Unsolicited Proposal

Written without any request. You identify the problem and convince the audience it is worth solving; requiring strong persuasion.



Example: A startup proposes solar charging stations to a city authority.

Who is it for?

Internal Proposal

Written within the same organisation. The focus is on improvement or operational efficiency. The tone is less formal; shared context with the audience is assumed.



Example: An IT department proposes upgrading lab computers at a university.

External Proposal

Written for outside clients or organisations. More formal and structured. Must establish credibility and clearly communicate benefits to an unfamiliar audience.

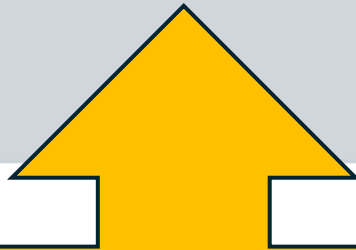


Example: A software company proposes a cybersecurity system to a bank.

Why is it written?

Planning Proposal

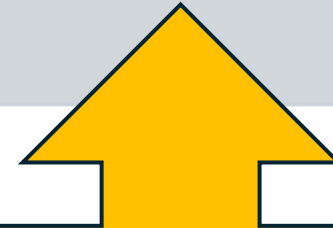
Focuses on how a project will be executed. Includes a step-by-step methodology, timeline, and resource plan.



Example: A CS team proposes a plan to develop a campus attendance management system, outlining phases, timeline, tools, and team roles.

Research Grant Proposal

Written to secure funding for academic or scientific research. Emphasises the research problem, methodology, and the significance of expected outcomes.



Example: A group proposes research on AI-based plagiarism detection, requesting funding and outlining objectives, methodology, and expected outcomes.

Why is it written?

Sales Proposal

Written to sell a product or service. Centres on client benefits, return on investment, and persuasive value propositions rather than technical detail.



Example: A tech company proposes a cloud-based cybersecurity system to a bank

Technical Proposal

How to write a Technical Proposal

Structure of a Technical Proposal



Background

Purpose

Set the scene before introducing the issue. Begin broadly, then narrow your focus to the relevant context.

Use a neutral, factual tone. Start broad, then slightly narrow your focus.

General situation



Current
practices/conditions



Relevant trends
(technology use,
demand etc.)



Background

Common Mistakes

Starting directly with the problem

Giving opinions instead of
context

Writing extensive detail too soon
in proposal

Statement of Problem: Identify the Gap

Purpose: Clearly identify the specific gap, who is affected, and the consequence of inaction.

Be Specific: Name the exact issue, not a general complaint

Show impact: Explain who is affected and what the consequences are

Stay Focused: Do not mix the problem with proposed solutions.



Show consequences

Statement of Problem



Be precise and direct, avoid vague language.

Common Mistakes

Writing vague problems

Repeating background instead of identifying a gap

Mixing problem with solution

Need: Why action is required?

Build on the problem: Why should we care?

Urgency



Write how the problem can worsen without intervention

Necessity



Link directly to measurable improvement

Logical reasoning



Build on the stated problem to justify a solution

Purpose: Justify why the problem must be solved now. Build logically on the problem statement to convey urgency



Avoid repeating
the content of
problem

Need: Why action is required?

Common Mistakes

Repeating the problem

Too general

Not showing urgency

Benefits: Value of the solution

Demonstrate concrete value

Focus on outcomes, not features

You may list two to three, specific, outcome-oriented advantages

Practical



Technical



Social/environmental



Limitations: Show Realism

Acknowledging limitations build credibility.

Reviewers trust proposals that are honest about constraints

Keep it balanced: neither dismissive nor overly negative

Cost and budget constraints



Maintenance and operational challenges



Technical or environmental limitations

Scope: Define boundaries

Purpose: Define the precise boundaries of your proposal: What is covered and,, what is not covered.

Within Scope

For example: You may include target area, location, or user group

For example: Specified systems or processes addressed

For example: Defined deliverables and coverage.

Out of Scope

For example: You may ass areas or systems not addressed

For example: Future phases or extensions

For example: Tasks or coverage outside the project.

Proposed Plan: Implementation

What to include:

What will be done?



How it will be implemented?

Basic steps or
phases

Purpose: Explain how the solution will be carried out in practice.